

Constructor University

Spring Semester 2026

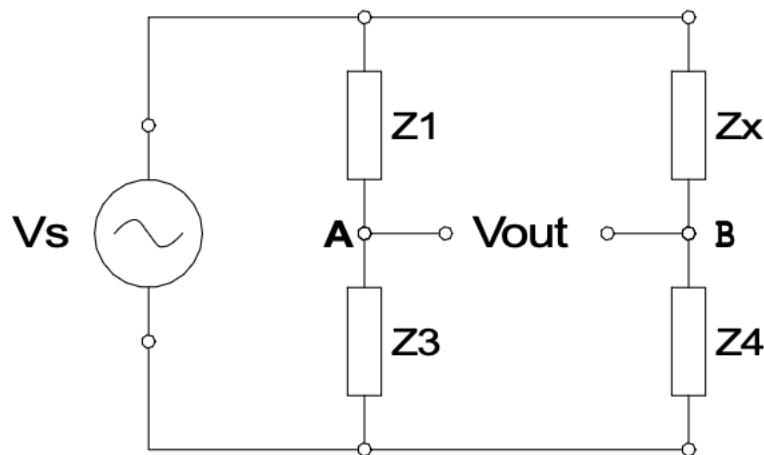
Lab Experiment 3: The Wheatstone Bridge

Author of the report: MR. Giorgi Kipshidze

Experiment conducted by : Giorgi Kipshidze, Jung Woojin
Place of execution : Teaching Lab EE
Date of execution : April 23 2026

1. Introduction

The Wheatstone Bridge is Electrical engineering circuit tool to measure unknown resistance or impedance with high accuracy. It consists of four resistive or impedance elements arranged in a bridge configuration and supplied by a voltage source.



Wheatstone bridge can be in two conditions: balanced and unbalanced. In the balanced, the output voltage between the two points is zero. In unbalanced the voltage in V_{out} is nonzero.

In a DC Wheatstone bridge, all elements are resistors. In an AC Wheatstone bridge, the elements may include capacitors and inductors, so impedance, phase shift, and frequency dependence must also be considered. Therefore, an AC bridge is more complex.

The aim of this experiment was to investigate balanced and unbalanced Wheatstone bridges in DC and AC circuits. The experiment also examined temperature sensor and the effect of measuring instruments on the bridge output voltage.

2. Execution

Part 1: Balanced DC Wheatstone Bridge

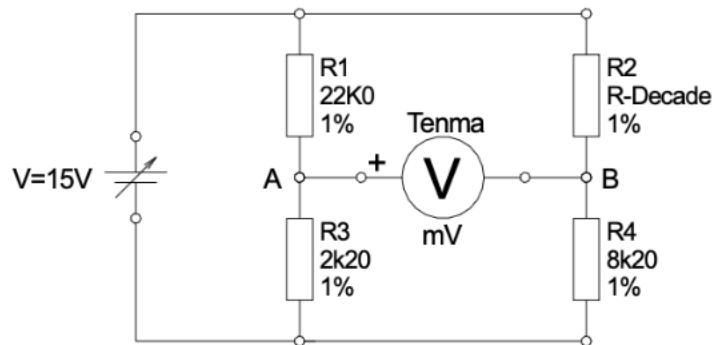
1.1 Experiment Setup

Workbench No.6

- Used tools and instruments
- Breadboard
- Resistor
- DC power supply
- Tenma multimeter
- Elabo multimeter

2.1 Part 1: Setup Balanced DC Wheatstone Bridge

A balanced DC Wheatstone bridge was constructed. The resistors R₁, R₃, and R₄ were measured using the Elabo multimeter.



Part 1 : Balanced DC Wheatstone bridge							
R1K Ω	R3 K Ω	R4 K Ω	Vab (mV)	R-decade (nominal value)	R-decade (real value)	R2 (ohm)	Elabo Multimeter (real value)
21.802	2.194	8.121	0	81.2k	81.54 k	81k 900R	15.001 V

1.3 Part 1: Balanced DC Wheatstone Bridge Execution and Results

The resistance of the decade box was adjusted until the voltage V_{ab} became as close to zero as possible. The Tenma multimeter was used in the mV range to measure the bridge output voltage.

$V_{ab}(mV0)$	R-decade(K Ω) real value	R-decade k Ω (nominal value)
0	81.54	81.2

Part 2: Unbalanced DC Wheatstone Bridge with PT1000

2.1 Experiment Setup

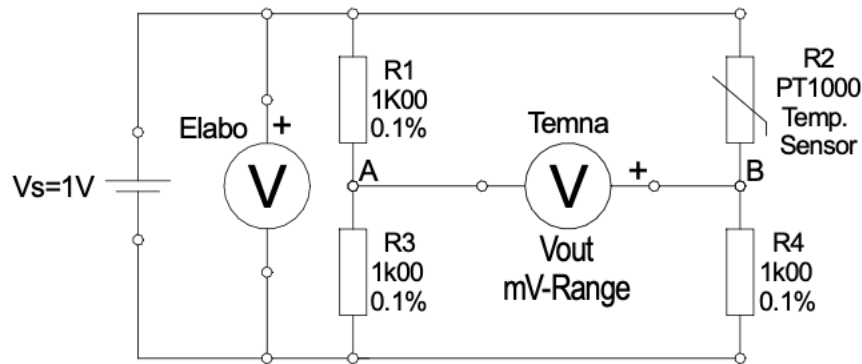
Workbench No.6

Used tools and instruments

- Breadboard
- Resistors
- PT1000 temperature sensor
- Tenma multimeter
- Elabo multimeter

2.2 Setup

An unbalanced Wheatstone bridge was built using a PT1000 temperature dependent sensor. The bridge output voltage was measured for two different supply voltages.



The PT1000 resistance changes with temperature according to: $R_T = R_0(1 + a\Delta T)$ where $R_0 = 1000 \text{ Ohm}$ and $a = 3.85 \times 10^{-3} \text{ C}^{-1}$

2.3 Results:

First, the power supply was set to 1 V and the bridge output voltage was measured.

Part 2 : Unbalanced DC Wheatstone bridge			
When $V_s = 1\text{V}$	V_s (real value)	V_{out} (mV)	
	1.0024	-22.9	

Then the voltage was set to 10 V, after waiting for 10 min, the voltage was measured again.

When $V_s = 10\text{ V}$	V_s (real value)	V_{out} (mV)
	10,006	-247.9

The output voltage increased strongly when the supply voltage was increased. So we can tell that unbalanced bridge is more reactive to changing resistance.

Part 3: Balanced AC Wheatstone Bridge

3.1 Experiment Setup

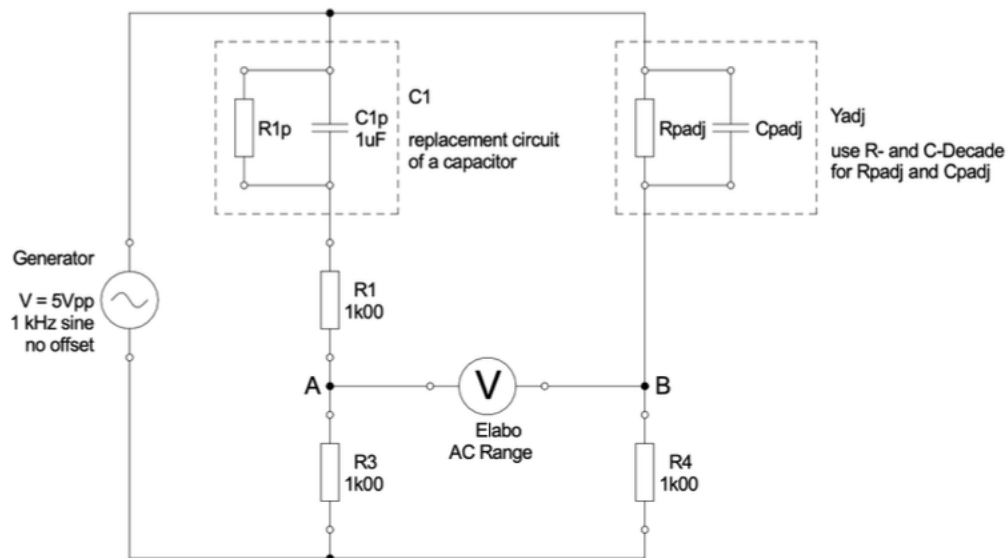
Workbench No.6

Used tools and instruments

- Breadboard
- Resistors
- Capacitor
- Resistor decade box
- Capacitor decade box
- Function generator
- Oscilloscope
- RLC meter
- Tenma multimeter

3.2 Setup

An AC Wheatstone bridge was constructed. The oscilloscope was used to observe the amplitude and phase shift between the two bridge points. Channel 1 was connected to point A and Channel 2 to point B.



Part 3 : Balanced AC Wheatstone bridge			
R1 (kohm)	R3	R4	C1 (nF)
0.9919	0.993	0.9919	944.92

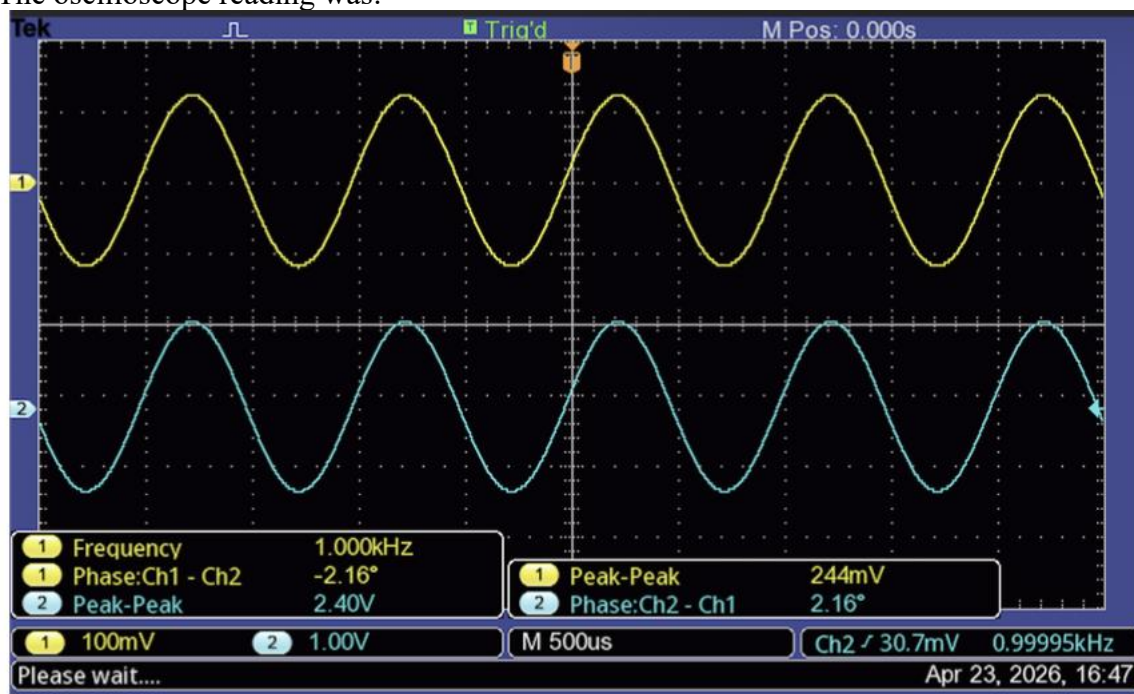
The bridge was supplied with an AC signal of approximately 5 V peak-to-peak at 1 kHz.

3.3: Results

The adjustable resistance R_{padj} and capacitance C_{padj} were changed until the output voltage V_{AB} was as close to zero as possible. A value below 5 mV would have been ideal, but the recorded value was 11 mV.

Vab	Radj	Cadj	Radj (T)	Cadj (T)
11 mV	1.015	27 nF	1.03 k ohm	25 nF

The oscilloscope reading was:



	Value
V_{PP}	2.4
Amplitude	1.2
Phase	2.16

The phase difference was small, showing that the bridge was close to balance, but not perfectly balanced.

3. Evaluation

Part 1: Balanced DC Wheatstone Bridge - Evaluation

1. calculating R_2

For balanced bridge this condition must be true: $V_{AB} = 0$. so $V_A = V_B$

Using voltage divider: $V_A = V_S \frac{R_3}{R_1 + R_3}$ $V_B = V_S \frac{R_4}{R_2 + R_4}$ We get that $\frac{R_4}{R_2 + R_4} = \frac{R_3}{R_1 + R_3}$

$$R_3(R_2 + R_4) = R_4(R_1 + R_3) \quad R_3R_2 + R_3R_4 = R_1R_4 + R_3R_4 \quad R_3R_2 = R_1R_4$$

$$\text{Therefor: } R_2 = \frac{R_1 R_4}{R_3}$$

Nominal values are: $R_1 = 22000 \text{ ohm}$ $R_3 = 2200 \text{ ohm}$ $R_4 = 8200 \text{ ohm}$

$$R_2 = \frac{22000 \times 8200}{2200} = 82000 \text{ ohm}$$

Now we do same with measured values: $R_1 = 21802$ $R_3 = 2194$ $R_4 = 8121$

$$R_2 = \frac{21802 \times 8121}{2194} = 80699.2 \text{ ohm}$$

The measured values from decade box and Elabo multimeter were:

$$R_{2\text{decade}} = 81.2k \text{ ohm} \quad R_{2\text{elabo}} = 81.54k \text{ ohm}$$

Errors were calculated : $\frac{\Delta R_2}{R_2} = 1\%$ for decade box

Error for elabo: $\frac{\Delta R_2}{R_2} = 0.085\%$

Using partial differentiation, the maximum relative error is: $\frac{\Delta R_2}{R_2} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_4}{R_4} + \frac{\Delta R_3}{R_3} = 3\%$

For calculated values= 0.2535%

Comparison of R_2 value

Method	R_2	Maximum relative error
Read from decade box	81.2k ohm	1%
Directly measured with Elabo	81.54k ohm	0.09%
Calculated from nominal values	82.0k ohm	3%
Calculated from measured values	80.7k ohm	0.25%

The best value is the one measured directly with the Elabo multimeter. The Wheatstone bridge result was less accurate mainly because the resistors used in the bridge had tolerances of about 1%

Part 2: Unbalanced DC Wheatstone Bridge - Evaluation

1. Calculation of PT1000 Resistance and Temperature

For $V=1$ case

$$V_s = 1.0024V \quad V_{out} = -22.9mV$$

The calculated resistance was $V_{pt} = 1095.96 \text{ ohm}$

Using: $R_T = (R_0(1 + aT)) \quad T = \frac{R_T - R_0}{R_0 a}$ after plugging values we get $T=24.9 \text{ C}$

For case 10 V :

$$V_s = 10.006V \quad V_{out} = 0247.9 \text{ mV} \quad R_{pt} = 1108.93 \text{ ohm} \quad T = \frac{1108.93 - 1000}{1000 \times 0.00385} = 28.3C$$

The measured temperature was higher in the 10 V case because the larger supply voltage caused more power dissipation in the PT1000 sensor.

	V_s	V_{out}	R_{PT1000}	Temperature
Case 1	1.0024V	-22.9mV	1095.96 ohm	24.9C
Case 2	10.006V	-247.9mV	1108.93 ohm	28.3C

2. Self-Heating of the PT1000 Sensor and multimeter consideration

When current flows through the PT1000 sensor, power is dissipated in it. This causes the sensor to heat up. The temperature increase due to self-heating is calculated using:

$$\Delta T = E \times P \text{ where } E = 0.2c/cW$$

For 1V case: $\Delta T = 0.05C$. *Very small value, no effect on measurement*

For 10V case: $\Delta T = 4.99C$. *This is significant increase*. Therefore, the 10 V measurement gave a higher temperature mainly because the PT1000 was heated by its own power dissipation.

This shows that a lower supply voltage is better when using temperature sensors in a Wheatstone bridge.

Now the influence of the voltmeter is considered. The Tenma multimeter was used in the mV range. In this range, its internal resistance is approximately: $R_{multi} = 2.5G \text{ ohm}$

This resistance is very large, so almost no current flows through the multimeter. Therefore, the bridge output voltage is not changed much.

If a voltmeter with only $10k\Omega$ internal resistance was used, the voltmeter would load the bridge. This means that current would flow through the voltmeter, and the measured output voltage would change.

The relative error for a $10k\Omega$ voltmeter was calculated as approximately: 9.24%

This is a large error. Therefore, the conclusion is that the output voltage of a Wheatstone bridge should always be measured with a voltmeter that has a very high input resistance.

Part 3: Balanced AC Wheatstone Bridge – Evaluation

In Part 3, the objective was to balance an AC Wheatstone bridge. In a DC bridge, only resistance values must be balanced. But in an AC bridge, the impedances have both magnitude and phase.

Because of this, one adjustable component is not enough. A resistor changes mainly the amplitude, while a capacitor changes the phase and the amplitude. Therefore, two adjustable components are needed: R_{padj} and C_{padj} and if balanced $V_{AB} = 0$

In AC case, phase and amplitude must be balanced: $R_1 = 991.9\Omega$ $R_3 = 993\Omega$ $R_4 = 991.9\Omega$ $C_1 = 1\mu F$

The frequency was: $f = 1kHz$ $\omega = 6283.1rad/s$

The capacitance C_1 $X_c = \frac{1}{\omega C_1} = \frac{1}{6283.19 \times 1\mu F} = 1.59.15\Omega$

For adjustable resistor: $R_{padj} = \frac{R_4}{R_3} \times \frac{R_1^2 + X_c^2}{R_1} = 1.016k\Omega$

For adjustable capacitor: $C_{padj} = \frac{R_3}{R_4} \times \frac{C_1}{1 + (\omega R_1 C_1)^2} = 25.1nF$

So theoretical values are: $R_{padj} = 1.016k\Omega$ $C_{padj} = 25.1nF$

The measured adjusted values: $R_{padj} = 1.015k\Omega$ $C_{padj} = 27nF$

The measured bridge output voltage was: $V_{AB} = 11mV$

The calculated and measured values are close to each other. The resistor value is almost the same. The capacitance value is slightly different, but still close.

Now the complex conductance is calculated. Using the measured values of R_1 , R_3 , R_4 , and Y_{adj} , the conductance was found to be: $Y_1 = (-0.174 + j5.880)mS$

Capacitance was calculated $C_{1p} = 0.936\mu F$ and parallel resistance was $R_{1p} = -5.74k\Omega$

We get negative resistance so something is wrong and result is not fully reliable.

RLC measured values $C_{1p} = 1.093\mu F$ and $R_{1p} = 37.534k\Omega$

Quantity	Calculated value	Measured value
C _{1p}	0.936 μ F	1.093 μ F

R_1p	-5.74k ohm	37.534k ohm
------	------------	-------------

The capacitance result is not perfect, but it is still acceptable for this experiment. The resistance result is not acceptable because it is negative. This shows that the AC bridge was not perfectly balanced.

The voltmeter and oscilloscope also affect the bridge. The voltmeter has input resistance, so small current flows through it. This changes the bridge output slightly. The oscilloscope has both input resistance and input capacitance. In an AC circuit, the input capacitance can change both amplitude and phase.

The conclusion for Part 3 is that an AC Wheatstone bridge is more difficult to balance than a DC Wheatstone bridge. In DC, only the resistance ratio matters. In AC, both amplitude and phase must be adjusted. This is why both R_{padj} and C_{padj} are needed.

4. conclusion

In this experiment, the Wheatstone bridge was investigated in balanced DC, unbalanced DC, and balanced AC configurations.

In the balanced DC Wheatstone bridge, the unknown resistance was determined using the bridge balance condition. The measured decade-box value was 81.2k ohm, while the direct Elabo measurement was 81.54k ohm. The calculated value from nominal resistors was 82.0k ohm, and the value calculated from measured resistors was 80.7k ohm. The most accurate result was obtained from the Elabo multimeter because it had the smallest relative error.

In the unbalanced DC Wheatstone bridge, a PT1000 sensor was used to measure temperature. The bridge output voltage changed when the PT1000 resistance changed. At 1 V supply voltage, the calculated temperature was about 24.9 C. At 10 V, the calculated temperature increased to about 28.3 C. This increase was mainly caused by self-heating of the PT1000 sensor. The self-heating was small at 1 V but became bigger at 10 V.

In the AC Wheatstone bridge, both resistance and capacitance had to be adjusted to balance the circuit. This is because AC impedances have both magnitude and phase. The measured adjustable values were close to the theoretical values, but the bridge was not perfectly balanced. The calculated capacitance had an error of about 14.4%, and the calculated parallel resistance was negative so it could be real. This shows that AC bridge measurements are more sensitive to phase errors, and the influence of measuring instruments.

Overall, the experiment showed that the Wheatstone bridge is a useful circuit for measuring resistance, impedance, and sensor changes. However, accurate results require precise components, careful balancing, low-tolerance resistors, and measuring instruments with high input impedance.

5. Reference

Lab manual GEE II

Elabo Multimeter - Accuracy